

Step 1: Full Supervised PLS Model

This model intentionally uses the full predictor universe before filtering obvious inverse or redundant variables. Mean turnout (outcome_mean_participation_citizen_18plus) supervises the latent components through PLS.

Model Fit

- Observations: 583
- Predictors: 70
- Selected components by 10-fold CV: 12
- Training R2: 0.693
- Cross-validated R2: 0.565
- Cross-validated RMSE: 0.057

Main Read

The full model is useful as a stress test. It lets the supervised dimension reduction procedure see every available signal, but it also lets inverse and same-concept variables double-enter the component construction. VIP scores should therefore be read as supervised importance, not as final variable-selection decisions.

The selected components are latent variables, not original predictors. In this model, 12 components means PLS built 12 turnout-supervised dimensions from the 70 original predictors. The original variables are interpreted through VIP scores, coefficients, and component loadings.

Highest VIP Variables

Variable	Family	VIP	Turnout corr	PLS direction
block3_visible_minority_share	immigration_citizenship	2.286	-0.642	lower turnout
block3_immigrant_share	immigration_citizenship	2.031	-0.573	lower turnout
block1_bachelors_or_higher_25_64_share	household_class	1.985	0.554	higher turnout
block1_unemployment_rate_share	household_class	1.934	-0.532	lower turnout
block4_effective_mayoral_candidates_5pct	mayoral_competitiveness	1.907	-0.535	lower turnout
block1_average_household_size	household_class	1.834	-0.485	lower turnout
block4_mayoral_vote_fragmentation	mayoral_competitiveness	1.775	-0.495	higher turnout
block3_non_official_mother_tongue_share	immigration_citizenship	1.751	-0.491	higher turnout
block3_non_citizen_share	immigration_citizenship	1.534	-0.417	lower turnout
block3_citizen_adult_share	immigration_citizenship	1.401	0.353	lower turnout
block1_median_age	age_structure	1.268	0.253	lower turnout
block3_recent_immigrant_share	immigration_citizenship	1.227	-0.317	lower turnout
block1_age_65_plus_share	age_structure	1.219	0.213	higher turnout
block4_mayoral_top_two_margin	mayoral_competitiveness	1.151	0.288	lower turnout
block4_mayoral_winner_margin	mayoral_competitiveness	1.151	0.288	lower turnout
block4_federal_margin	federal_competitiveness	1.065	0.272	higher turnout
block2_apartment_lt5_storeys_count	housing_form_density	1.034	0.275	higher turnout
block5_library_count_1200m	service_access	0.990	0.246	lower turnout
block3_english_french_knowledge_share	immigration_citizenship	0.969	0.266	higher turnout
block5_ksi_collision_events_2021_2025_per_1000	service_contact	0.933	0.160	higher turnout

Reference Categories Suggested By The Full Model

The full model points toward these broad reference categories for story-building: immigration and racialized geography, education and class resources, household composition, local electoral fragmentation, citizenship and language context, age structure, and urban service/contact geography. The strongest plain-language variables behind these categories are visible minority share, immigrant share, bachelor or higher education share, unemployment rate, effective mayoral candidates above five percent, average household size, mayoral vote fragmentation, non official mother tongue share, non citizen share, citizen adult share, median age, recent immigrant share.

Short Interpretation

Variables with VIP above 1 are contributing more than average to the turnout-supervised PLS projection. The strongest variables should be carried into Step 2 and Step 3 as evidence, but not accepted mechanically. For example, if renter and owner variables both score highly, that is not two separate housing-tenure findings; it is the same tenure concept appearing twice with opposite coding.